PROJECT: Custom VPC setup with Bastion Host & NAT Gateway

Services: VPC, EC2, NAT Gateway, Bastion Host, Security Groups.
Project Goal: Build a secure private/public subnet architecture with internet access for private instances.
Skills: Routing Tables, Subnetting, SSH Tunnelling.

Definition: A custom VPC setup with a Bastion Host and NAT Gateway is an AWS network architecture where a user defined Virtual Private Cloud is created with public and private subnets, a Bastion Host is deployed in the public subnet to provide secure administrative access to private instances, and a NAT Gateway is used to allow private instances outbound internet access without exposing them to inbound internet traffic.

Scope of the Project: The scope of this project is to design and implement a secure and scalable AWS network architecture using a custom Virtual Private Cloud (VPC) that separates resources into public and private subnets.
The project focuses on providing:

- Secure administrative access to provide EC2 instance using a Bastion Host.
- Outbound internet connectivity for private instances using a NAT Gateway.
- Strong network isolation and access control using security groups and route tables.

Methodology:

1. Create VPC
2. Create 2 subnets:
   - Public
   - Private
3. Attach Internet Gateway to public subnet
4. Create NAT Gateway for private subnet
5. Launch:
   - Bastion in public
   - Private EC2 in private

Services Used in this Project:

1.**VPC (Virtual Private Cloud)**
Service: Amazon VPC
- Creates your own private network
- You define IP range (CIDR)

2. Subnets

Part of VPC

- Public Subnet → internet access
- Private Subnet → no direct internet

3. EC2 (Virtual Servers)

Service: **Amazon EC2**

- Bastion Host (in public subnet)
- Private Server (in private subnet)

4. Internet Gateway

Service: **Internet Gateway**

- Allows public subnet to connect to internet

5. NAT Gateway

Service: **NAT Gateway**

- Allows private servers to access internet
- Blocks incoming internet traffic

6. Route Tables

Service: **VPC Route Tables**

- Controls traffic flow

Example:

- Public → IGW
- Private → NAT

7. Security Groups

Service: **Security Groups**

- Controls inbound & outbound traffic

8. Key Pair

Service: **EC2 Key Pair**

- Used to login (SSH) to instances

9. Elastic IP

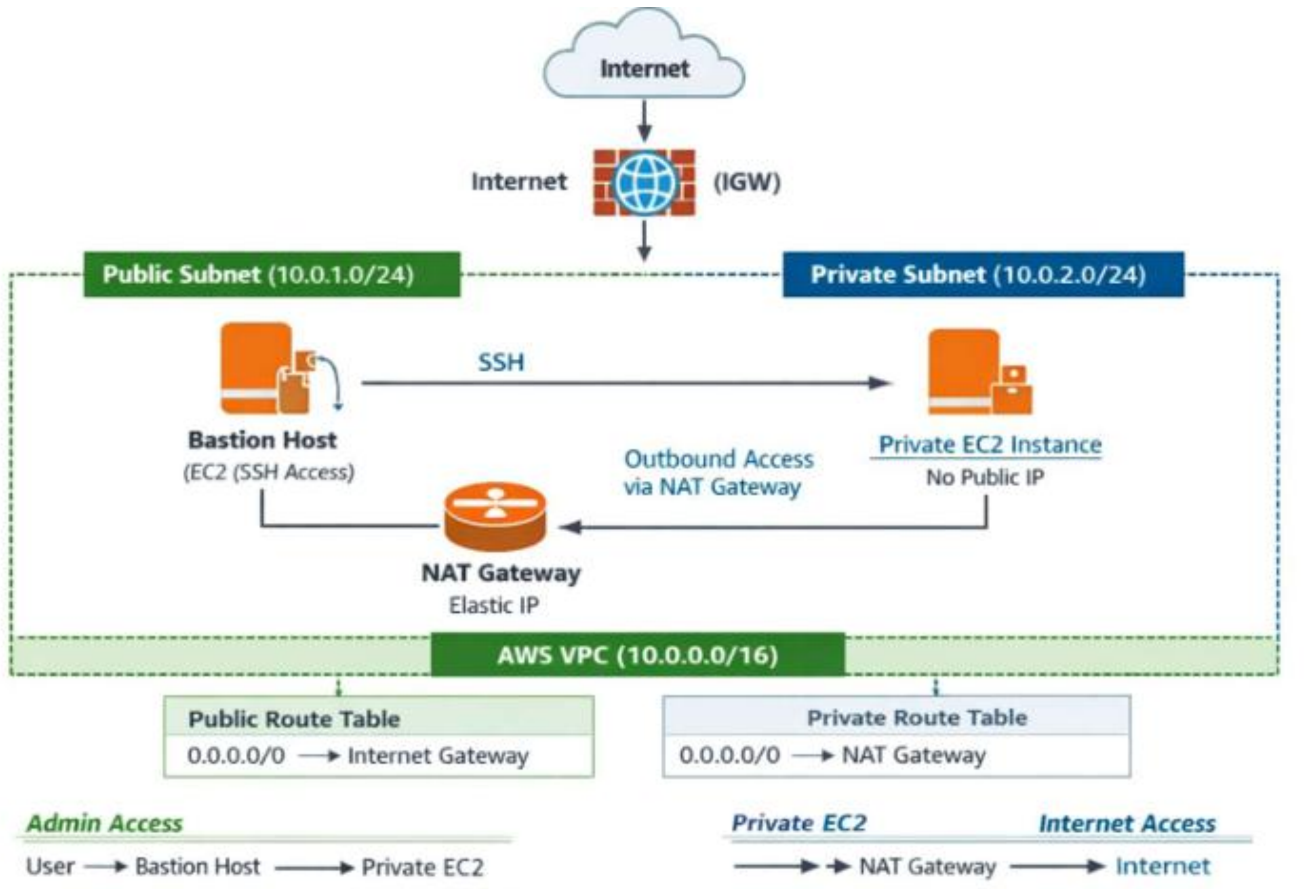
Service: **Elastic IP**

- Used for:
  - Bastion Host
  - NAT Gateway

10. SSH (Secure Shell)

Entity: Secure Shell (SSH)

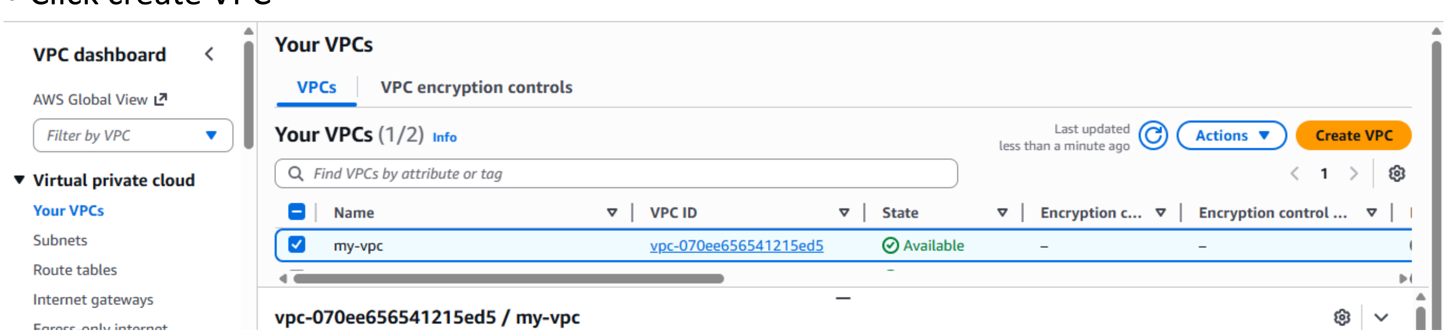
Architecture Diagram:



Implementation and Execution of a custom VPC setup with Bastion Host and NAT Gateway.

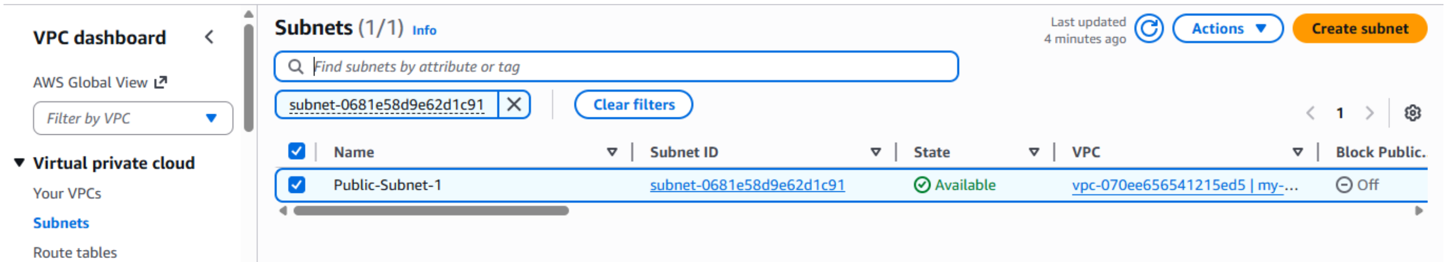
Phase 1: Create VPC.

- Open AWS Management Console
- Click create VPC
- Select VPC only
- Configure: -Name: my-vpc-IPV4 CIDR: 10.0.0.0/16 -Tenancy: Default
- Click create VPC



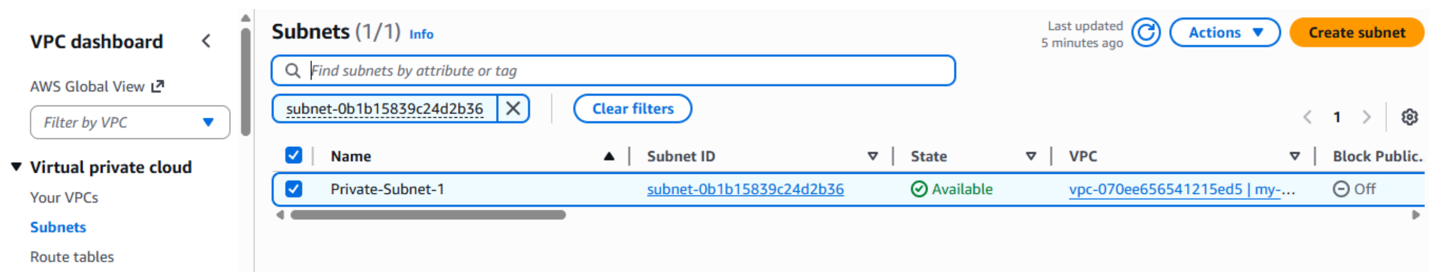
Phase 2: Subnet:

- Go to Subnets --Create Subnet
- Select VPC: my-vpc
- Subnet Name:Public-Subnet1
- Availability Zone: ap-south-1a
- IPV4 CIDR: 10.0.1.0/24
- Click create subnet.



Private Subnet:

- Go to Subnets--Create Subnet
- Subnet Name: Private-Subnet1
- Availability Zone: ap-south-1a
- IPV4 CIDR: 10.0.2.0/24
- Click create Subnet

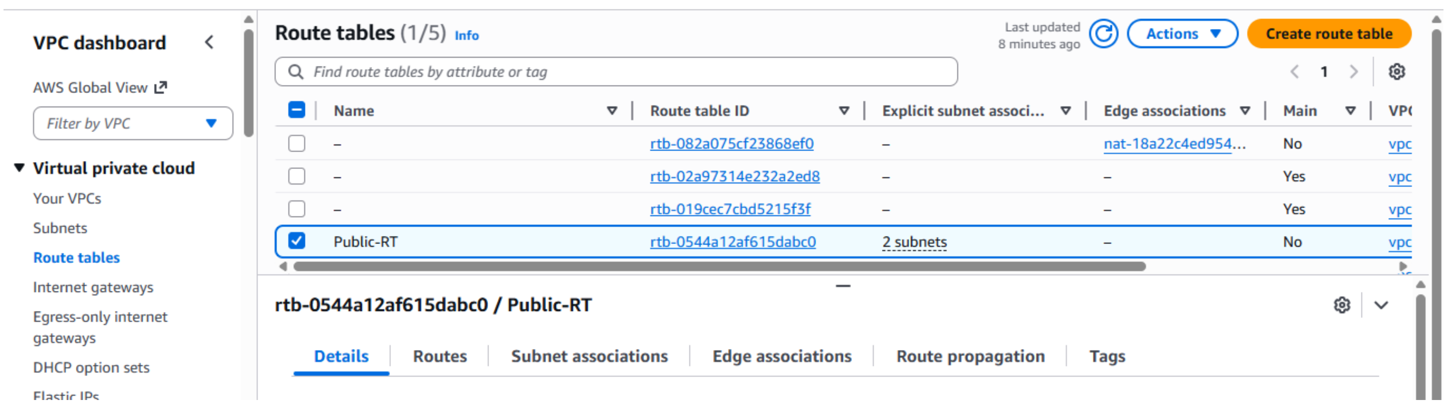


Phase 3: Create Internet Gateway (IGW)

- Go to Internet Gateway ⍰ Create
- Name: my-IGW
- Click create Internet Gateway
- Select IGW ⍰ Actions ⍰ Attach to VPC
- Choose Bastion-VPC
- Click on Attach.

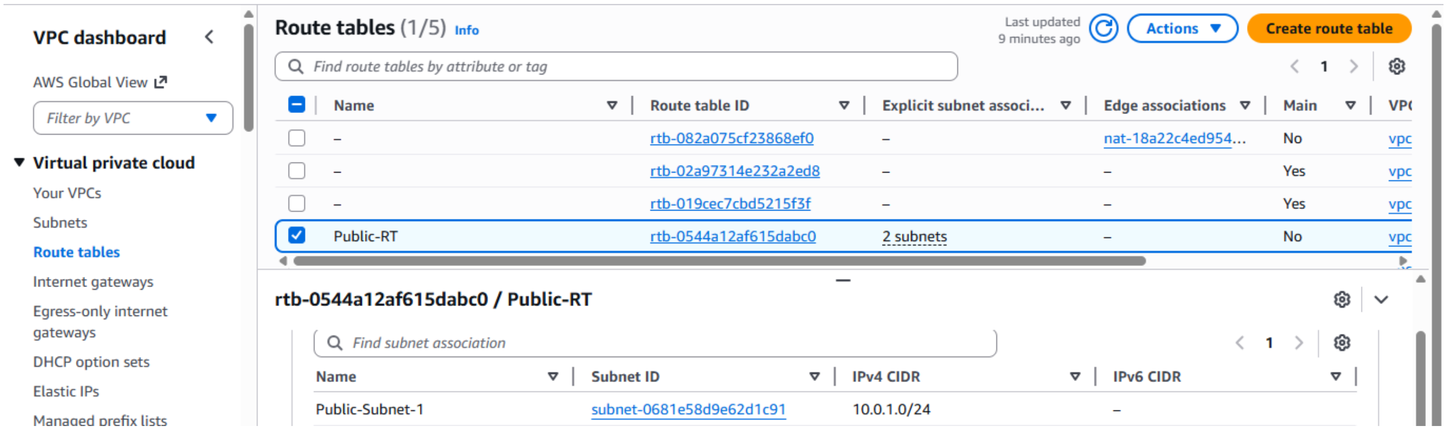
Phase 4: Create Public Route Table Public Route Table:

- Go to Route Tables---Create route table
- Name: Public-RT
- VPC: my-vpc
- Click create route table Add Route:
- Select Bastion-Public-RT1
- Routes: Edit routes
- Add routes: ¬ Destination: 0.0.0.0/0 ¬ Target: Internet Gateway (Bastion-IGW)
- Save routes



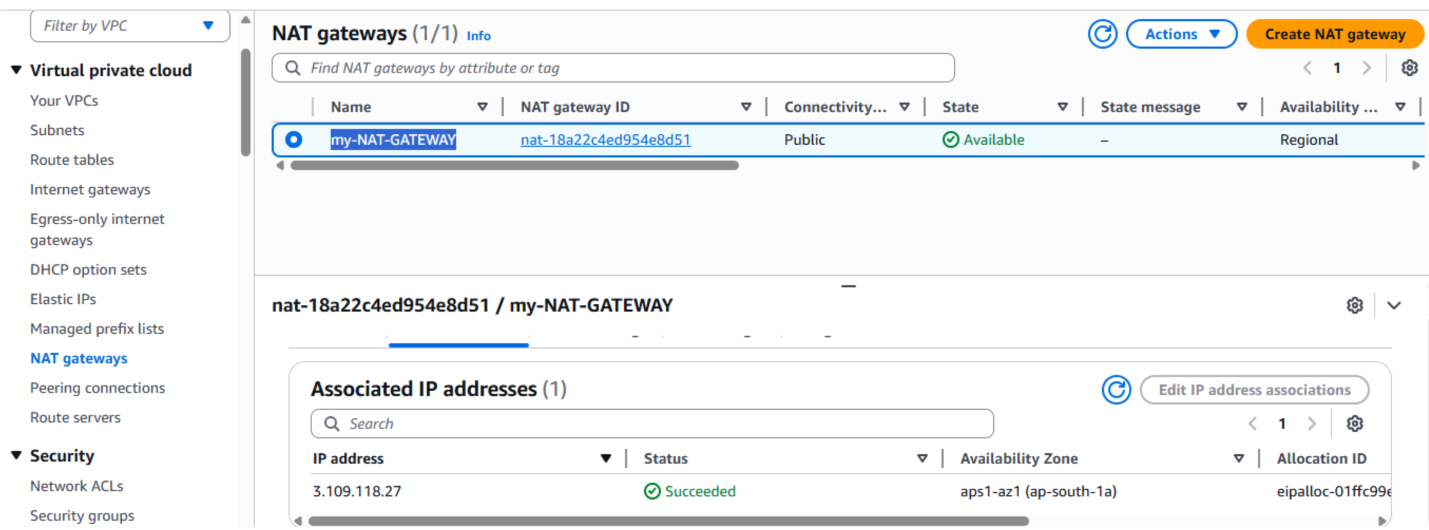
Associate Public Subnet:

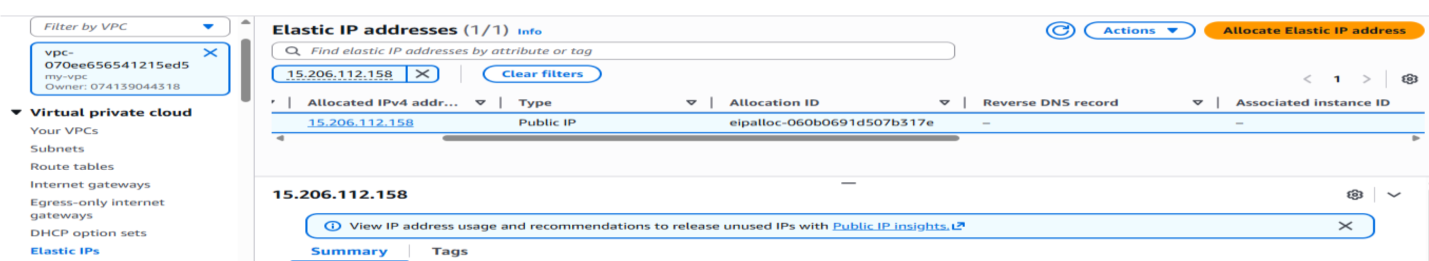
- Subnet Associations: Edit
- Select Public-Subnet1
- Save



Phase 5: Create NAT Gateway

- Go to NAT Gateway---Create
- Name: my-NAT-GATEWAY
- Subnet: Public-Subnet1
- Connectivity type: Public
- Elastic IP: Allocate Elastic IP
- Click create NAT Gateway
- Wait until status becomes Available





Phase 6: Create Private Route Table:

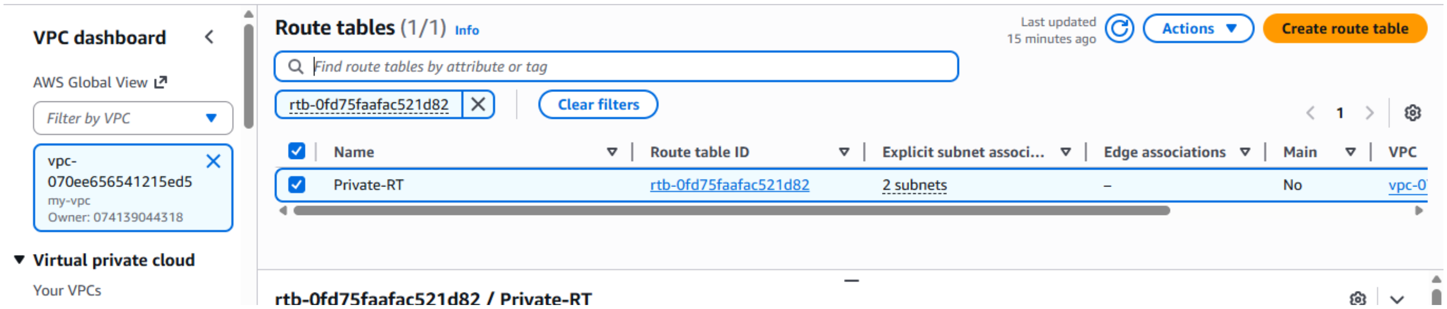
Private Route Table:

Go to Route Tables---Create

Name: Bastion-Private-RT1

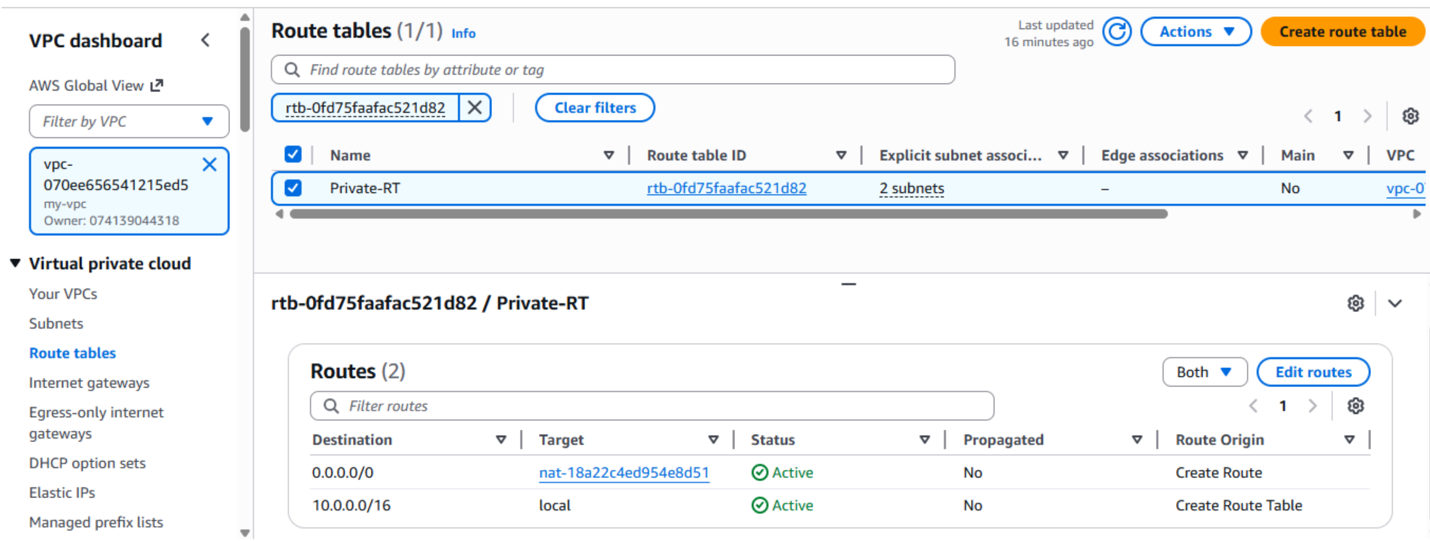
VPC: Bastion-VPC

Click create



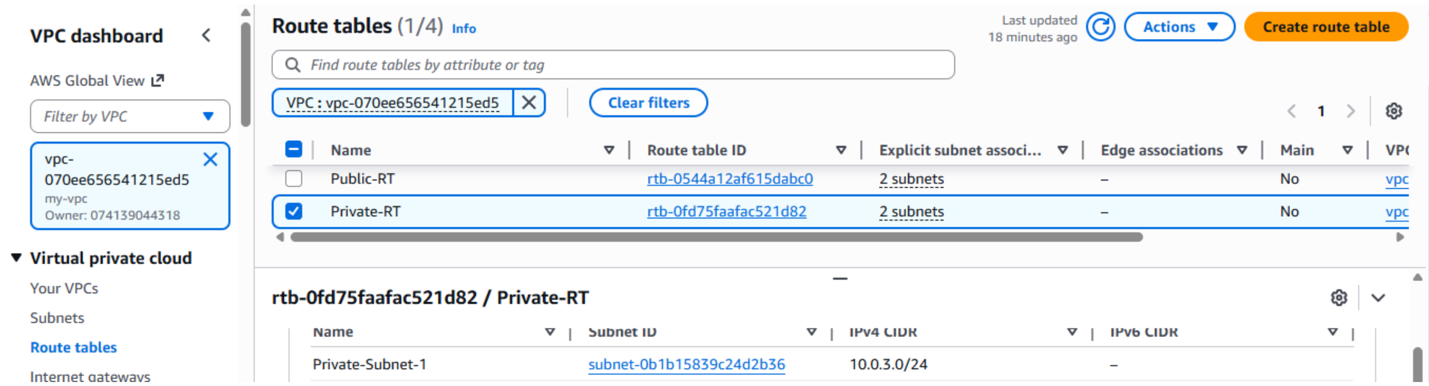
Add NAT Route:

- Select Bastion-Private-RT1
- Routes: Edit routes
- Add: ¬ Destination: 0.0.0.0/0 ¬ Target: NAT Gateway (Private NAT Gateway1)
- Save



Associate Private Subnet:

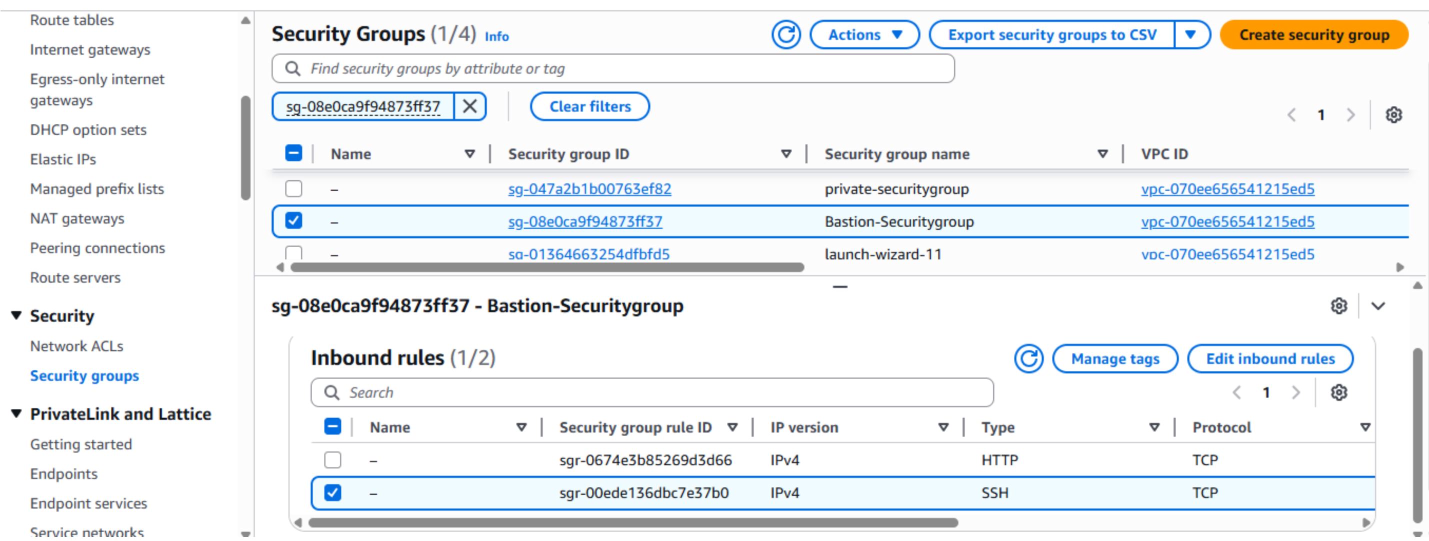
- Subnet associations: Edit
- Select: Private-Subnet-1
- Save



Phase 7: Create Security Group

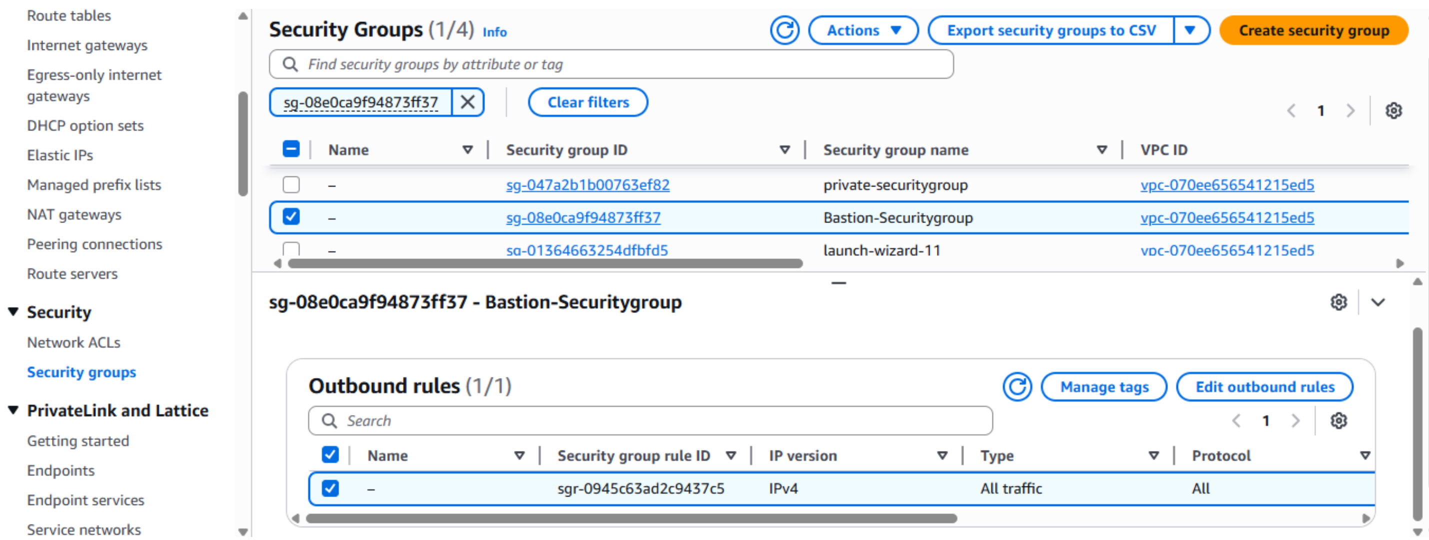
Bastion Host Security Group:

- Go to Security Group Create
- Name: Bastion-securitygroup
- VPC: Bastion-VPC
- Inbound Rule:
- Type: SSH
- Port Range: 22
- Source: Anywhere IPv4 (0.0.0.0)



- Outbound Rule:
  - Type: All traffic
  - Port Range: All

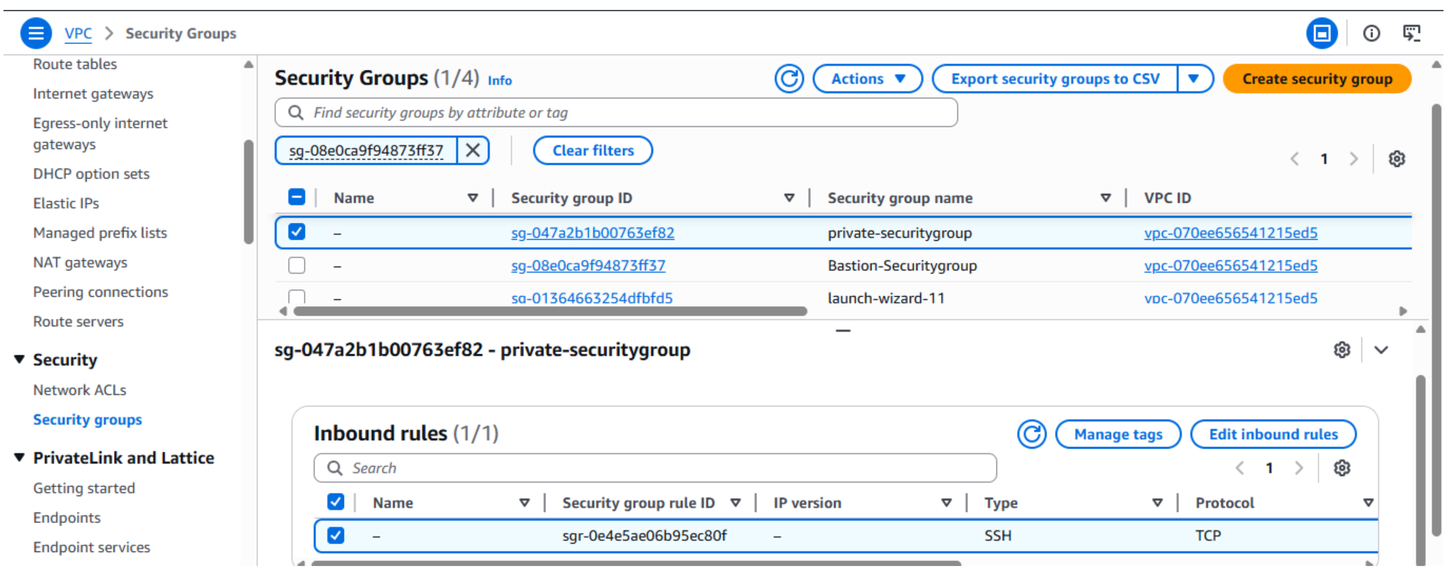
- Destination: 0.0.0.0/0
- Click create



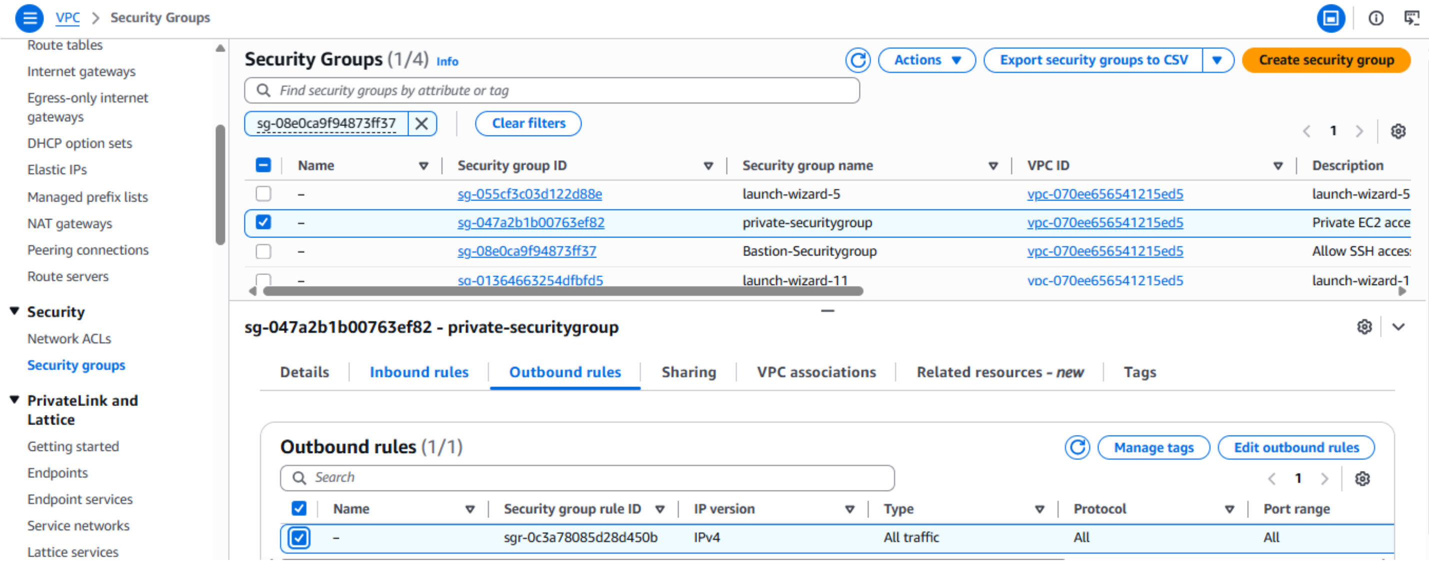
Phase 8: Create Private EC2 Security Group

Private EC2 Security Group:

- Create Security Group
- Name: Private-Securitygroup
- VPC: my-vpc
- Inbound Rule: ¬
- Type: SSH, All ICMP
- Source: Bastion-Securitygroup (Bastion-Host)

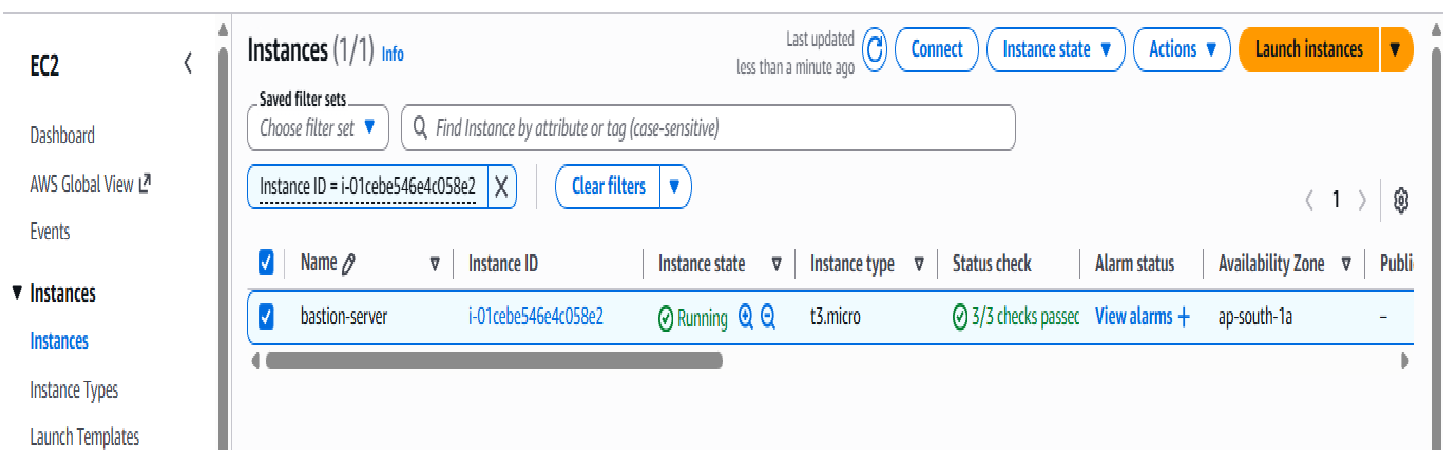


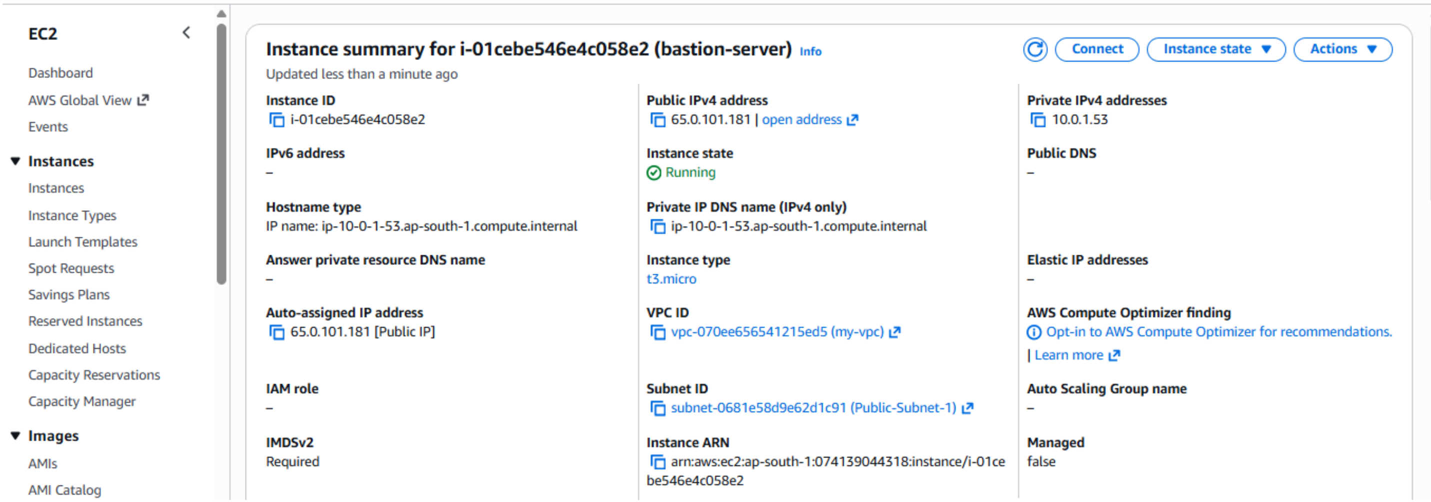
- Outbound Rule:
  - Type: All traffic
  - Destination:0.0.0.0/0
  - Click create



Phase 9: Launch Bastion Host (Public EC2 Instance)

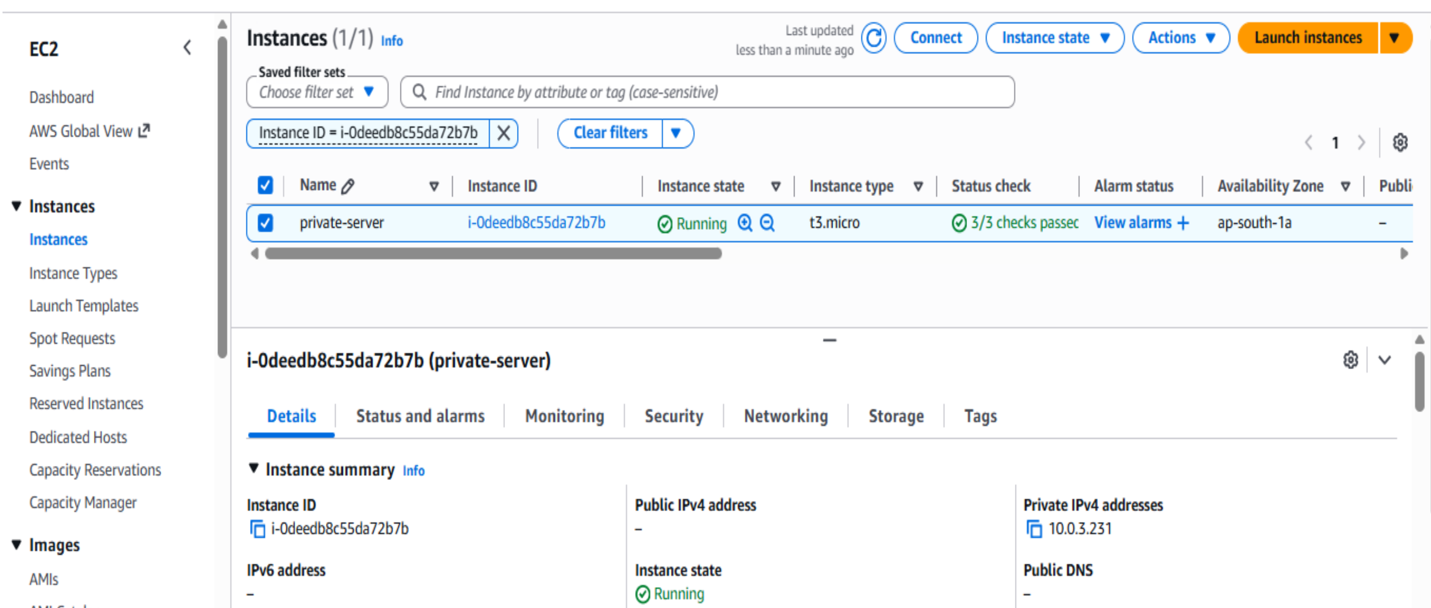
- Go to EC2---Launch instance
- Name: Bastion-Server
- AMI: Amazon Linux Kernel 6.1 2023
- Instance Type: t3. micro
- Key-Pair: Select/Create (Bastion-Key)
- Network Settings:
- VPC: my-vpc
- Subnet: Public-Subnet-1-ap-south-1a
- Auto-assign Public IP: Enable
- Security Group: Create Bastion-Securitygroup
- Launch instance

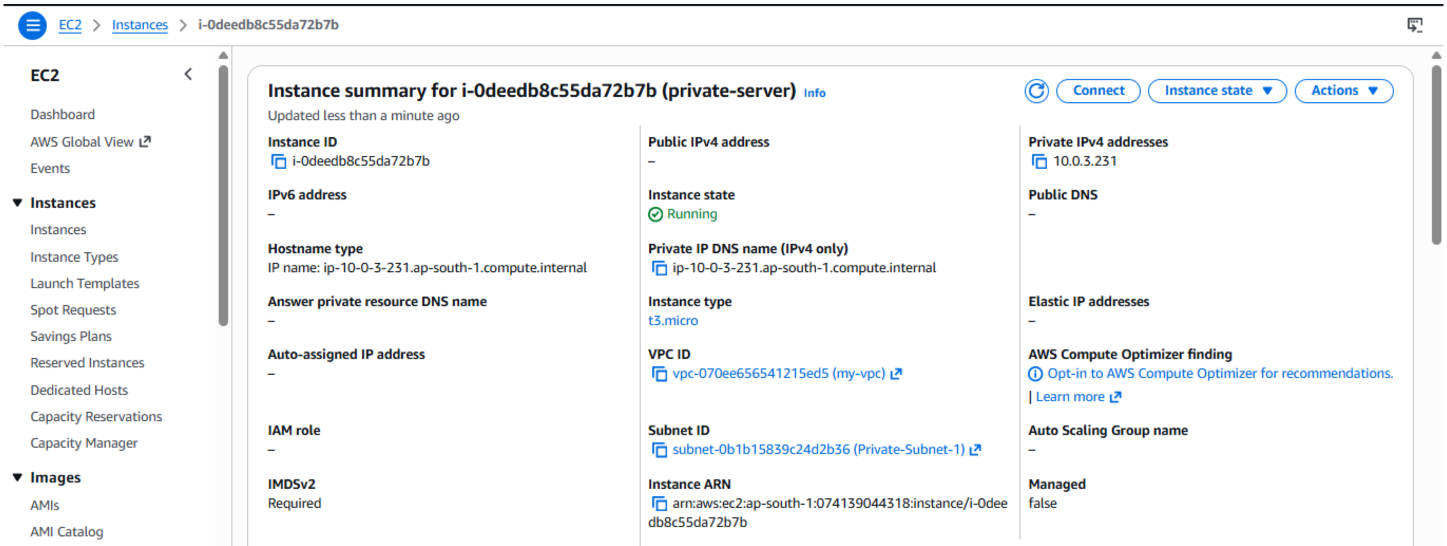




Phase 10: Launch Bastion Host (Public EC2 Instance)

- Go to EC2---Launch instance
- Name: Private-Server
- AMI: Amazon Linux Kernel 6.1 2023
- Instance Type: t3. micro
- Key-Pair: Select/Create (Bastion-Key)
- Network Settings:
- VPC: my-vpc
- Subnet: Private-Subnet-1-ap-south-1a
- Auto-assign Public IP: Disable
- Security Group: Create Private-Securitygroup
- Launch instance



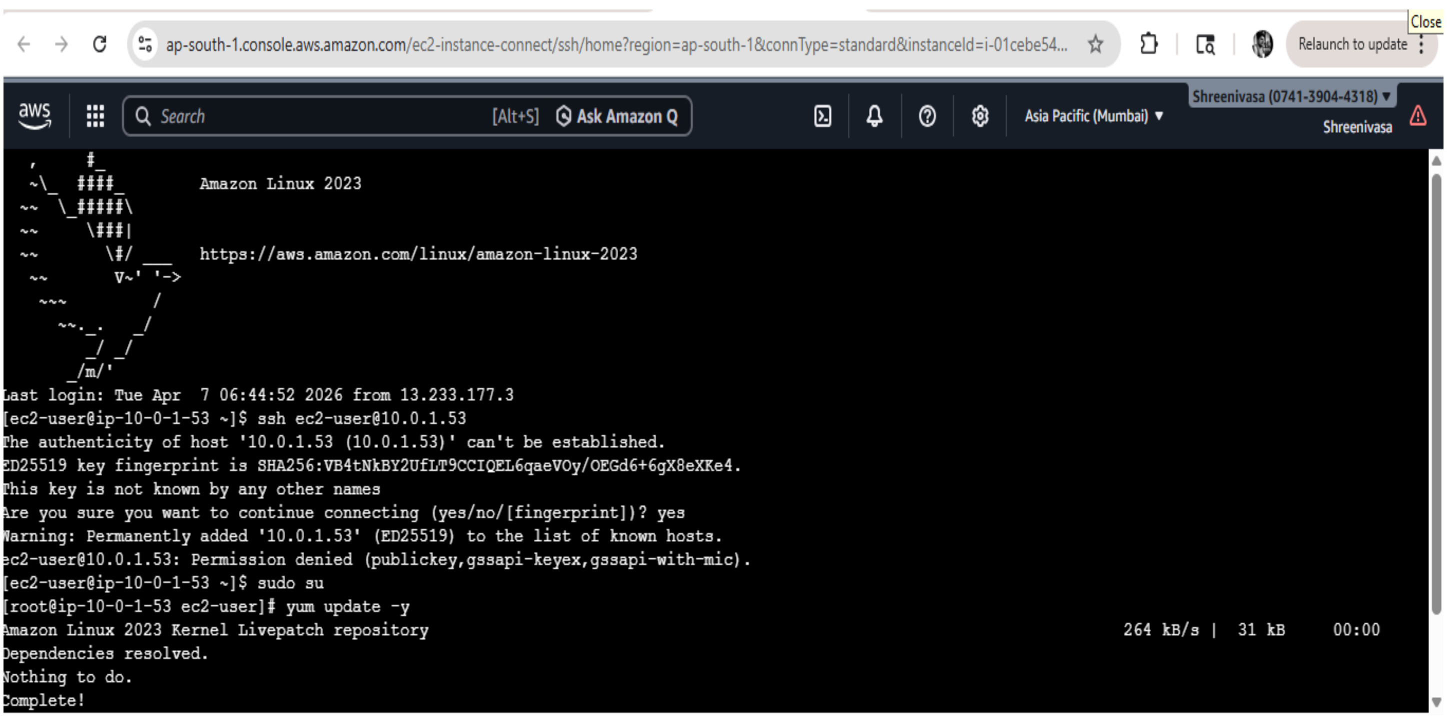


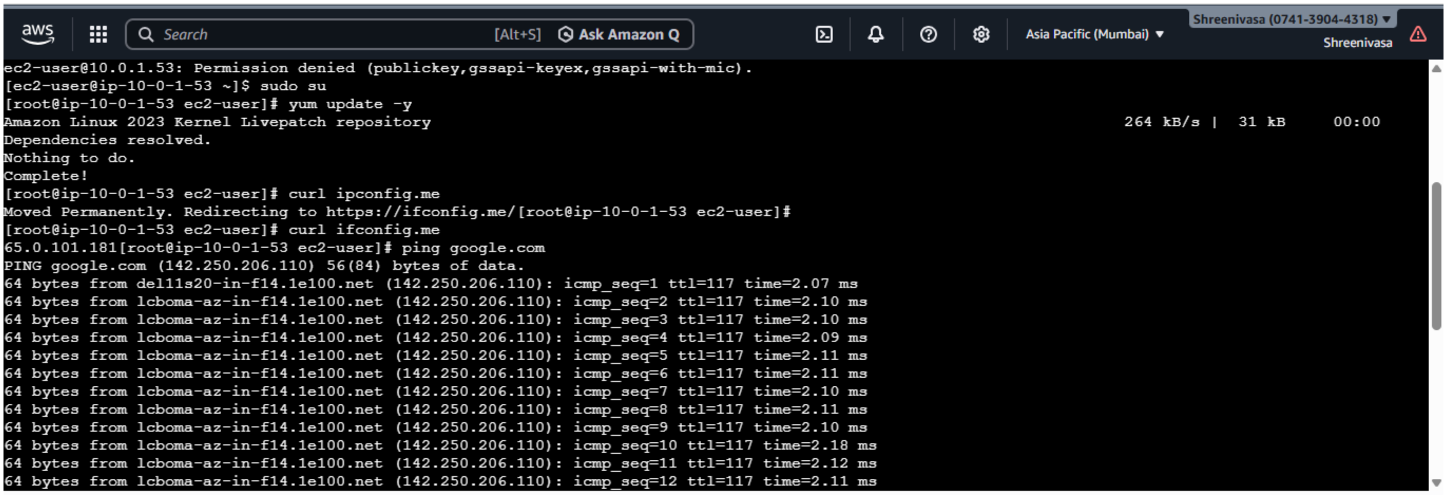
Phase 11: SSH Access (Bastion Host to Private Ec2 instance)

- From Local Machine:

Used Command Prompt

- From Bastion-Host:





Troubleshooting:

In this custom VPC project for me ssh was created in wrong way instead of giving source type Anywhere IPv4 I gave myIP its not working .

Solution: later I changed to Anywhere IPv4 in the source type. It worked properly.

Conclusion:

The custom VPC setup with Bastion Host and NAT Gateway successfully demonstrates the implementation of a secure and well-structured AWS network architecture. By segregating resources into public and private subnets, the project ensures strong network isolation and enhanced security. The Bastion Host acts as a controlled entry point for administrative access, while the NAT Gateway enables private EC2 instances to access the internet securely without being exposed to inbound traffic. Proper configuration of route tables and security groups ensures efficient traffic flow and adherence to AWS best practices. Overall, this project provides hands-on experience in designing, deploying, and managing a scalable and secure cloud infrastructure suitable for real-world production environments.